

MovieIQ — Predictive Analytics on Film Success

Machine Learning Project Brief • Predicting Whether a Movie Will Succeed

Language	Python	Libraries	Pandas, NumPy, scikit-learn, SciPy
Visualization	Seaborn, Matplotlib	App / Dataset	Streamlit · movies.csv

In this project you will build **MovieIQ**, an interactive dashboard that analyses and predicts the success of movies from key performance indicators — budget, revenue, popularity, runtime, and average votes. You will explore the data, run statistical tests, train a machine-learning model to predict success, and finally wrap everything in a deployable **Streamlit** app. A movie is considered **successful** when its revenue is greater than its budget. Work through the stages below and answer every question with code, visuals, and short written explanations.

***What to submit:** (1) your Python source (including the Streamlit app file), (2) a `requirements.txt`, (3) any charts saved to an `assets/` folder, and (4) short written answers to the questions below. If you deploy the app, include the live link. Every claim must be reproducible from your code.*

STAGE 0 Problem Statement

1. In your own words, explain what makes a movie “successful” in this project. Write the exact rule you will use to label a movie as a success or a failure.
2. Why is predicting film success valuable? Name two stakeholders (e.g. studios, investors) and how such a prediction could help them.
3. State the **objective** of the project and list at least three concrete steps you will take to reach it.
4. This is a classification problem. Explain what that means and what your model's **target variable** will be.

STAGE 1 Data Preparation

Your dataset (`movies.csv`) should contain these columns:

```
budget, revenue, popularity, runtime, vote_average, title, genres
```

1. Load the dataset. How many rows and columns does it have? Print summary statistics for the numeric fields.
2. Check for **missing values** and **zeros** in budget and revenue. Why can a budget or revenue of 0 be a problem, and how will you handle these rows?
3. Create the **target column** `success = 1` when `revenue > budget`, else 0. What proportion of movies are successful? Is the dataset balanced?
4. The genres column often holds multiple genres per movie. How will you process it so it can be used for filtering and analysis?

STAGE 2 Exploratory Data Analysis

1. Plot **Budget vs. Revenue** using a scatter plot. Describe the relationship. Do higher budgets tend to earn higher revenue?
2. Explore **genre trends**: which genres are most common, and which tend to be most successful? Visualise your answer.
3. Examine how **popularity, runtime, and vote_average** relate to success. Which of these looks most associated with a successful movie?
4. Produce a **correlation heatmap** of the numeric features. Which pairs are strongly correlated, and does that raise any concerns for modelling?

STAGE 3 Statistical Testing

1. Run a **T-Test** to check whether a numeric feature (e.g. popularity or vote_average) differs significantly between successful and unsuccessful movies. State your null hypothesis, the p-value, and your conclusion.
2. Run a **Chi-Square test** to check whether a categorical feature (e.g. genre) is associated with success. State the hypothesis, result, and what it means.
3. In plain language, what does a **p-value** tell you? What threshold did you use to decide significance, and why?

STAGE 4 Predictive Modeling (Random Forest)

1. Select your **features** and target. Which columns did you feed the model, and why did you exclude any (e.g. title, or revenue itself)?
2. Split the data into **training and test** sets. What split ratio did you use, and why is a separate test set important?
3. Train a **Random Forest Classifier**. Briefly explain, in your own words, how a random forest makes a prediction.
4. Evaluate the model using accuracy, precision, recall, and a confusion matrix. How well does it predict success? Where does it make the most mistakes?
5. Inspect **feature importance**. Which features matter most for predicting success? Does this agree with your EDA and statistical tests?

STAGE 5 Streamlit Dashboard & Deployment

Wrap your analysis into an interactive, deployable app.

1. Build a Streamlit dashboard that lets a user **filter by genre and by minimum vote average** via the sidebar, and updates the visuals accordingly.
2. Display your key EDA charts and statistical-test results inside the app.
3. Add a section where a user can input a movie's details and receive a **success / not-success prediction** from your trained model.
4. Write a `requirements.txt` and run the app locally with `streamlit run MovieIQ.py`. Confirm it launches cleanly.
5. **(Bonus)** Deploy the app publicly (e.g. Streamlit Community Cloud) and include the live link. What did you have to change for deployment to work?

***Reflection:** If a studio asked you "Will our next film succeed?", how confident would you be in MovieIQ's answer? Name one limitation of your data or model, and one thing you would improve with more time.*